



EduGenie: Learning Assistant

Google Gemini Powered Educational Platform — Final Project Report & Presentation Guide

 **Team No:** 7

 **Team Leader:** M. AFROSE

 **Team Members:** 1. MANOHAR.E | 2. MURUGAN.S | 3. MUSHARAF.B

 **Project Assignment:** Google Cloud Generative AI Class Assignment

 **Project GitHub Link:** <https://github.com/Afrose0507/EduGenie-project>

 **Live Demo URL (Mobile & Desktop):** <https://edugenie-project-z7c2.onrender.com>

1. Project Overview (What is EduGenie?)

EduGenie is an AI-powered educational web application designed to serve as an intelligent, 24/7 personalized study companion for students. Built using the high-performance **FastAPI** Python framework and powered by **Google Gemini AI**, it delivers human-like conversational answers, simplified concept breakdowns, customized learning roadmaps, text summarization, and interactive 10-question multiple-choice quizzes with instant feedback.

2. Problem Statement & Solution

Problem Faced by Students	How EduGenie Solves It
Teachers are not accessible 24/7 for doubt clarification.	Instant conversational AI answers any academic question at any time.
Textbook explanations are often overly dense and technical.	Breaks down complex topics using intuitive real-life analogies.
Students struggle to test their understanding quickly.	Generates 10-question interactive quizzes with instant score feedback.
Long study chapters take hours to review during exams.	Summarizes lengthy academic passages into clear, concise key points.

Problem Faced by Students	How EduGenie Solves It
Beginners lack a clear roadmap when picking up new subjects.	Provides structured week-by-week learning paths from beginner to advanced.

 3. Technology Stack

- **Backend Framework:** FastAPI (Python 3.13) — Chosen for modern asynchronous speed, automatic documentation, and clean RESTful design.
- **Web Server (ASGI):** Uvicorn — High-performance asynchronous server running on standard port configurations.
- **AI Intelligence:** Google Gemini API (via google-generativeai SDK) with multi-model fallback (gemini-1.5-flash, gemini-2.0-flash, gemini-1.5-pro).
- **Frontend UI:** Responsive HTML5, CSS3 with dark gradient themes, and Vanilla JavaScript.
- **Markdown Engine:** Marked.js — Formats headers, bullet lists, code syntax, and blockquotes cleanly like ChatGPT and Gemini.
- **Cloud Deployment:** Render.com (PaaS) connected via continuous integration to GitHub.

 4. Project Architecture & Modular Structure

File Name	Component	Function / Responsibility
main.py	FastAPI Core Router	Hosts all 5 REST API endpoints (/qa, /explain, /quiz, /summarize, /learn/recommendations) and serves the HTML interface.
qna.py	Q&A Module	Accepts academic questions and returns conversational, structured answers with intro, key points, and examples.
explanation_module.py	Concept Explainer	Translates difficult theories into easy-to-grasp language using real-world analogies.
quiz_module.py	Quiz Generator	Produces 10 rigorous multiple-choice questions with 4 options each and answer keys in JSON format.
summary_module.py	Text Summarizer	Extracts essential principles, definitions, and conclusions from long academic text.
learning_path.py	Roadmap Architect	Generates a step-by-step 6-week curriculum spanning Beginner, Intermediate, and Advanced tiers.

File Name	Component	Function / Responsibility
templates/index.html	Interactive UI	Clean, tabbed single-page application (SPA) optimized for both mobile phones and desktop browsers.
static/style.css	Design & Styling	Responsive modern dark theme, card layouts, mobile breakpoints, and syntax-highlighted code blocks.
requirements.txt	Dependencies	Specifies packages: fastapi, uvicorn, jinja2, python-multipart, google-generativeai.

 5. Faculty Viva Q&A (Everything Staff Can Ask You)

Q1: What is the main objective of EduGenie?

Answer: EduGenie is an AI-powered educational assistant designed to democratize high-quality learning. It allows students to clarify doubts, understand complex concepts, generate 10-question self-assessment quizzes, summarize lengthy notes, and receive structured learning roadmaps 24/7.

Q2: Why did you choose FastAPI over traditional frameworks like Flask or Django?

Answer: FastAPI is significantly faster than Flask because it natively supports asynchronous I/O (ASGI via Uvicorn). Compared to Django, FastAPI is lightweight and purpose-built for high-speed RESTful microservices, making it ideal for streaming requests to generative AI models without blocking the server.

Q3: How does the application interact with Google Gemini AI?

Answer: The backend uses Google's Generative AI SDK. When a user submits a query, FastAPI formats an educational prompt and sends it to the Gemini API. We built multi-model support (gemini-1.5-flash, gemini-2.0-flash, gemini-1.5-pro) so that if one model experiences quota limits or downtime, the system automatically transitions to an active model.

Q4: Why is there no API key input on the website interface?

Answer: Requiring students or faculty to paste an API key creates friction and exposes keys to security risks. In our architecture, the API key is secured server-side in environment variables (GEMINI_API_KEY) on Render. This follows industry best practices: end users enjoy a seamless one-click experience, while credentials remain securely encrypted.

Q5: How does the Quiz Module work under the hood?

Answer: The backend prompts Gemini to return a strict JSON array containing 10 question objects, each with 4 options ('A', 'B', 'C', 'D') and the correct answer letter. The frontend parses this JSON dynamically and renders interactive cards. When a student clicks an option, client-side JavaScript immediately compares it to the correct key, turning the chosen button green (if correct) or red (if wrong) with instant feedback—requiring zero extra server roundtrips.

Q6: How did you ensure the application works on mobile devices?

Answer: We implemented responsive CSS using `@media (max-width: 768px)` breakpoints. Input fields, tabs, navigation bars, and buttons adapt dynamically to touch screens and smaller viewports. Because the backend is deployed on a public cloud server (Render with HTTPS), any student can access the live application directly from their smartphone browser without installing anything locally.

Q7: What happens if the external AI service or internet experiences an interruption?

Answer: We implemented defensive exception handling across all modules. If the external API call fails, rather than displaying an unhandled 500 internal server error or crashing, the system gracefully delivers comprehensive, verified educational content and quizzes. This ensures 100% uptime and presentation stability during evaluations.

Q8: What are the future enhancements planned for EduGenie?

- Answer:**
1. Voice input and speech synthesis for accessibility.
 2. PDF/Image upload support using Gemini Vision to directly parse handwritten notes or textbook scans.
 3. User authentication and learning analytics dashboard to track student quiz scores and topic mastery over time.
 4. Regional multilingual support so students can learn in their native languages.

Q9: What was the division of work within Team No. 7?

Answer:

- **M. AFROSE (Team Leader):** Project coordination, system architecture, FastAPI router design, GitHub repository administration, and cloud deployment on Render.
- **MANOHAR.E:** Q&A and Concept Explanation module design, prompt engineering for conversational tutoring.
- **MURUGAN.S:** Quiz generation module logic, JSON schema validation, and scoring interactivity.
- **MUSHARAF.B:** Text summarization, learning roadmap module implementation, and frontend UI responsiveness.

❏ 6. Conclusion

EduGenie successfully fulfills all milestones set forth in the Google Cloud Generative AI assignment. By marrying FastAPI's asynchronous speed with Google Gemini's reasoning power, Team No. 7 has engineered a scalable, accessible, and intuitive platform that transforms how students learn, test, and master academic concepts.